E.7: Six Systems, One Constraint: Design Lessons from Algorithmic Alternative Representation

Gio Della-Libera

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Abstract

Algorithmic redistricting research has largely focused on optimizing the drawing of single-member districts. This paper reports what we learn about representation design more broadly by comparing six algorithmically implemented alternatives to the standard single-member geographic district: multi-member districts (E.1), county-based representation (E.2), national redistricting without state lines (E.3), partisan-similarity clustering (E.4), partisan-fairness seat allocation (E.5), and international applications (E.6). Five design lessons emerge from systematic comparison. First, geography is destiny for single-member districts: the Rodden sorting gap produces systematic underrepresentation of Democrats regardless of the drawing method, algorithmic or otherwise. Second, multi-member districts partially but not fully escape geographic sorting: three-member districts reduce the gap by approximately 40%, five-member by 65%. Third, eliminating districts entirely (county representation) trades proportionality in competitive states for a fractional-voting mechanism that is institutionally unfamiliar and constitutionally problematic. Fourth, crossing state lines produces geographically compact but politically fragmented districts: the national map achieves 12% better mean Polsby–Popper compactness but preserves 40% fewer counties. Fifth, the compactness–proportionality Pareto frontier is steep: gaining one percentage point of proportionality costs approximately 0.015 Polsby–Popper units, and no feasible single-election geographic system escapes this constraint. These lessons imply a design hierarchy: for single-member systems, the constraint hierarchy (population $\rightarrow$ contiguity $\rightarrow$ compactness) is optimal under geographic sorting; multi-member districts are the only structural escape available within current constitutional constraints.

1 Introduction: The Meta-Question

What do we learn about representation design by implementing six alternative systems and comparing them algorithmically? This is the meta-question that animates the E-track research program and that this paper addresses as its capstone.

The six papers in Track E — multi-member districts [Della-Libera, 2026b], county representation [Della-Libera, 2026c], national redistricting [Della-Libera, 2026d], partisan-similarity clustering [Della-Libera, 2026e], partisan-fairness allocation [Della-Libera, 2026f], and international applications [Della-Libera, 2026g] — each ask a local question: what happens to compactness, proportionality, minority representation, or county preservation when we change this one design dimension? The synthesis paper E.0 [Della-Libera, 2026a] organizes these local questions into a comparative Pareto analysis. This paper asks the deeper question: what general principles about representation system design do we extract from the systematic comparison?

Our framing is borrowed from engineering design theory, where the concept of the *design space* refers to the full set of possible configurations of a system, and the *Pareto frontier* refers to the set of configurations from which no improvement on one dimension is possible without degrading another [Simon, 1996]. We argue that redistricting is a design problem in exactly this sense: there is a well-defined space of possible representation systems, a small number of key performance dimensions, and a set of constraint surfaces that bound the achievable region. The E-track experiments are controlled explorations of this space.

1.1 Why Algorithmic Comparison Enables Design Learning

The conventional approach to comparing representation systems is political science field research: examine real-world systems (US single-member districts, German mixed-member proportional, Irish STV, etc.) and compare outcomes. This approach has fundamental limitations. Real-world systems differ on multiple dimensions simultaneously — electoral formula, district size, ballot structure, candidate selection, campaign finance — making it impossible to isolate the effect of any single design choice. They also differ in their historical, institutional, and demographic contexts, introducing confounds that threaten causal inference [Lijphart, 1999].

The algorithmic approach overcomes these limitations by holding the algorithm constant and varying design dimensions one at a time. We use the same edge-weighted recursive bisection procedure [Karypis and Kumar, 1998, Della-Libera, 2026h] for all six experiments. The algorithm accepts the same census geography inputs in all cases; what changes is the objective function (single-member versus multi-member), the jurisdictional constraints (state-based versus national), the edge-weighting scheme (geographic versus partisan), and the seat-allocation mechanism (proportional versus winner-take-all). This controlled variation means that differences in outcomes can be attributed to design choices rather than to confounding contextual factors.

A second advantage is counterfactual reachability. Some alternatives — national redistricting, strict county representation — are constitutionally unavailable in the US but can be implemented computationally. The resulting maps are not real redistricting proposals; they are measurement instruments that answer the question "how much does the constitutional constraint cost us in this dimension?" Manual redistricting cannot answer this question because it cannot produce the counterfactual; algorithmic redistricting can.

1.2 The Five Lessons

We organize our findings into five design lessons, each addressing a specific dimension of the design space:

Lesson 1 (Section 2) Geography is destiny for single-member districts. The Rodden sorting gap [Rodden, 2019] is not a drawing artifact; it is a structural consequence of where Democratic voters live. No single-member geographic drawing method eliminates it.

Lesson 2 (Section 3) Multi-member districts partially but not fully escape geographic sorting. Larger districts average over more geographic heterogeneity, reducing the wasted-vote problem by 40% (three-member) to 65% (five-member), but the residual gap from extreme urban concentration persists.

Lesson 3 (Section 4) Eliminating districts entirely trades proportionality for local accountability. County representation is more proportional in competitive states but less responsive to population change and institutionally unfamiliar as a voting mechanism.

Lesson 4 (Section 5) Crossing state lines produces geographically compact but politically fragmented districts. The national map gains 12% compactness but loses 40% of whole-county preservation and eliminates state-based representation entirely.

Lesson 5 (Section 6) The compactness–proportionality Pareto frontier is steep. Gaining 1 pp of proportionality costs approximately 0.015 PP units. No system escapes this trade-off within a geographic single-election framework.

1.3 Scope and Method

All quantitative results in this paper are derived from the redistricting pipeline described in Della-Libera [2026h], applied to 2020 Census geography for all 50 states and 435 congressional districts. Political data (vote shares, efficiency gap, partisan bias) are computed from the 2016–2022 average of US House election results, following the methodology in Stephanopoulos and McGhee [2015]. County preservation rates are computed from Census TIGER 2020 boundary files. Where results differ from the companion papers E.1–E.6 (due to data vintage or parameter differences), we note this explicitly.

The paper targets the audience of *Annual Review of Political Science*: scholars familiar with redistricting debates and electoral system theory who seek a synthesizing analysis rather than the primary experimental findings. Readers seeking methodological detail should consult the companion papers.

2 Lesson 1: Geography Is Destiny for Single-Member Districts

The single most important finding from the E-track research program is also its most sobering: for any system that uses geographically compact single-member districts, Democratic underrepresentation relative to national vote share is a structural inevitability, not a drawing artifact. This lesson is established most directly by Papers E.4 and E.5, which approach the sorting problem from opposite directions.

2.1 The Rodden Mechanism

Rodden [2019] identifies the mechanism precisely. Democratic voters are geographically concentrated in dense urban cores in ways that produce "wasted votes" under winner-take-all single-member elections. A district centered on central Detroit, Philadelphia, or San Francisco will vote 85–15 Democratic regardless of how its boundaries are drawn; the 85% who vote Democratic "waste" their votes in the sense that the district would have produced a Democratic representative with 51% of the vote, and the additional 34 percentage points of Democratic margin are electorally inert. Republican voters, by contrast, are spread across suburban and rural geographies where their margins are more modest — close enough to win, but not so large as to generate systematic waste.

The result is that in a system with perfect population equality, Democratic voters systematically produce fewer Democratic seats per Democratic vote than Republican voters produce Republican seats. Chen and Rodden [2013] showed that this effect is large enough to produce Republican seat majorities in many states even when district lines are drawn by nonpartisan algorithms with no partisan intent. Our 50-state algorithmic sweep confirms this finding with more recent data and higher geographic resolution [Della-Libera, 2026h].

2.2 Evidence from E.4: Partisan-Similarity Clustering

Paper E.4 provides the sharpest demonstration that sorting is a structural rather than drawing artifact [Della-Libera, 2026e]. By varying the partisan-similarity edge weight from zero (pure compactness, the baseline) to a maximum that drives all partisan-similar tracts into the same district, E.4 shows that the aggregate seat–vote relationship remains almost unchanged.

Specifically, when we weight tract adjacencies by partisan similarity and optimize for partisan homogeneity within districts, the resulting maps have 95% safe seats (versus 52% in the baseline) and a much wider distribution of partisan margins. But the mean seat shares for both parties are nearly identical to the baseline: the algorithm has changed which seats each party wins safely and which it wins narrowly, but it has not changed how many seats each party wins in expectation. The seat bonus for Republicans — the gap between Republican seat share and Republican vote share — is +4.2 percentage points in the baseline and +4.0 percentage points in the partisan-similarity extreme, a difference of 0.2 pp that is not statistically distinguishable from zero ($p = 0.41$).

This null result for partisan redistribution within a single-member geographic system is the most direct evidence that sorting, not drawing, is the operative constraint.

2.3 Evidence from E.5: Partisan-Fairness Allocation

Paper E.5 approaches the same problem from the other direction: instead of asking what happens when we optimize for partisan homogeneity, it asks what the algorithm must do to achieve proportionality [Della-Libera, 2026f]. The answer is that it must draw a specific class of non-compact districts — those that crack urban Democratic concentrations into multiple surrounding suburban districts, reducing the margin wasted in each.

The quantity and severity of these non-compact districts is a direct measure of the geometric cost of escaping sorting. In states with severe Democratic urban concentration (Illinois, New York, Pennsylvania, Michigan), achieving proportionality requires districts that would fail any reasonable compactness threshold. In states with more evenly distributed partisanship (Iowa, Wisconsin, Arizona), the compactness cost of proportionality is modest.

This cross-state variation in the geographic cost of proportionality is itself informative. It implies that state-by-state adoption of proportionality standards would have uneven effects: states with less sorting would achieve proportionality at low compactness cost, while states with high urban concentration would face a much steeper trade-off. A national proportionality standard would thus impose differential burdens on different states.

2.4 The Implication: Drawing Methods Are Not the Remedy

The combined evidence from E.4 and E.5 establishes that improving the drawing method — whether by algorithmic means, independent commission, or court supervision — cannot eliminate the seat–vote gap arising from geographic sorting. The gap exists at the map level, not the drawing-process level. Courts that focus on drawing *process* (whether legislators had partisan intent) rather than drawing *outcomes* (whether the resulting map is proportional) are, by this analysis, addressing the wrong question for the wrong reason.

This finding does not imply that drawing-process reforms are without value. Algorithmic redistricting and independent commissions do eliminate the *additional* seat bias that partisan gerrymanders layer on top of the structural sorting bias. In some states, that additional bias is large (North Carolina, Ohio, Wisconsin in the 2010s) and its elimination is consequential. But in states where the structural sorting bias already dominates, no process reform changes outcomes materially.

3 Lesson 2: Multi-Member Districts Partially but Not Fully Escape Geographic Sorting

If single-member geographic districts cannot escape the Rodden sorting gap, can multi-member districts? Paper E.1 provides a detailed answer: partially, but not fully, and the degree of escape increases with district size in a predictable way [Della-Libera, 2026b].

3.1 The Mechanism of Partial Escape

The mechanism is straightforward. In a single-member district centered on an urban core, a 70–30 Democratic advantage produces one Democratic seat and zero Republican seats. In a three-member district centered on the same core with a surrounding suburban ring, the overall composition might shift to 60–40 Democratic, producing two Democratic seats and one Republican seat under proportional allocation. The margin improvement from 70–30 to 60–40 is modest, but the seat change from 1–0 to 2–1 is a 33% gain in Republican representation — a meaningful partial escape from sorting-driven underrepresentation.

The key variable is how much suburban and exurban population can be incorporated into an enlarged district without exceeding the population quota. Three-member districts have three times the population of single-member districts, so they reach further into suburban rings; five-member districts reach further still. The extent to which this averaging across geographic zones reduces partisan concentration depends on the city-suburb-rural gradient of the specific metropolitan area.

3.2 Empirical Evidence from E.1

Paper E.1 implements three-member and five-member districts using the same edge-weighted recursive bisection algorithm as the baseline but with population quotas scaled by district size [Della-Libera, 2026b]. The 50-state results show:

- **Three-member districts**: Seat–vote deviation reduced from the median sorting gap of +4.2 pp Republican (range: 1.1 pp in ND to 11.3 pp in WI; baseline) to +2.5 pp Republican, a 40% reduction in the sorting-driven gap. The gap reduction is smallest in low-sorting states (IA, ND) and largest in high-sorting states (IL, NY, PA, MI). Mean Polsby–Popper compactness declines from 0.367 to 0.286, a 22% reduction.

- **Five-member districts**: Seat–vote deviation reduced to +1.5 pp Republican, a 64% reduction. Mean Polsby–Popper compactness declines to 0.241, a 34% reduction.

The compactness decline is expected: larger districts must span more geographic territory and more heterogeneous terrain, producing less regular shapes. The important question is whether the proportionality improvement is worth the compactness cost.

3.3 Why Full Escape Is Impossible

The residual +1.5 pp gap that persists even with five-member districts reflects extreme urban concentration in a small number of metropolitan areas. Manhattan (New York County) votes approximately 85–15 Democratic in congressional elections; central Chicago similarly at 80–20; central Los Angeles at 75–25. These concentrations are so extreme that no realistically sized district can incorporate enough surrounding population to reduce the Democratic margin to proportional territory.

A five-member district centered on Manhattan with a population quota five times the single-member benchmark would encompass much of the outer boroughs, but the outer boroughs themselves are heavily Democratic, so the district-level composition remains far above proportional levels. The algorithm cannot extend the district into New Jersey or Connecticut without violating the state- boundary constraint.

The theoretical maximum reduction in the sorting gap from multi-member districting, holding state boundaries constant and allowing unlimited district size, is approximately 80% of the current gap. The remaining 20% is irreducible given the intensity of concentration in the largest American cities. This bound is derived from the fraction of Democratic votes that are in census tracts with $> 70\%$ Democratic composition — the "captive" Democratic votes that no district boundary can dilute.

3.4 The Size–Proportionality Trade-off

The data suggest a concave relationship across the three tested configurations (single-member, 3-member, 5-member districts), consistent with diminishing returns to increasing district magnitude: the marginal gap reduction is approximately 40% for the first additional two members (one to three) and approximately 24% for the next two (three to five). With only three data points, the functional form cannot be confirmed, but the pattern is sufficient to support the policy conclusion that the Fair Representation Act's 3–5 member specification captures most of the available improvement. The marginal gain diminishes because the most easily diversifiable geographic zones are incorporated first — the suburban rings adjacent to urban cores — and what remains is the irreducible urban concentration.

This concavity has practical implications. The Fair Representation Act's choice of three-to-five-member districts is near the knee of the marginal-benefit curve: it captures most of the available proportionality improvement while keeping ballot complexity and district-size effects manageable. Larger districts (seven or more members) buy little additional proportionality at substantially higher institutional cost.

3.5 Minority Representation Benefits

Multi-member districts provide a second proportionality benefit that single-member districts cannot: minority representation that is proportional to minority population share rather than dependent on creating majority-minority districts. In a single-member system, the *Gingles* framework [U.S. Supreme Court, 1986] requires majority-minority districts as a VRA remedy, but these districts can only be created where minority populations are geographically concentrated enough to form a majority.

In a three-member district, a 35% minority population share translates to one minority seat if minority voters prefer the minority-group candidate and bloc-vote accordingly — without requiring a majority-minority geographic concentration. Paper E.1 finds that three-member districts produce approximately 18% more effective minority representation than the baseline single-member system when minority populations are moderately dispersed [Della-Libera, 2026b]. This benefit complements rather than duplicates the VRA majority-minority district guarantee.

4 Lesson 3: Eliminating Districts Entirely Trades Proportionality for Local Accountability

County-based representation (Paper E.2) eliminates the redistricting problem by replacing drawn districts with fixed jurisdictional units whose boundaries are established by state statute and not subject to manipulation by congressional legislatures [Della-Libera, 2026c]. The design lesson from this experiment is that the cure for gerrymandering that county representation offers comes with two distinct costs: reduced local accountability through the fractional-voting mechanism, and reduced responsiveness to population change over time.

4.1 The Design Logic of County Representation

County representation applies the Huntington–Hill apportionment method — the same method used to allocate House seats among states since 1941 [Balinski and Young, 2001] — to allocate House seats among the 3,143 US counties and county-equivalents. A county with population equal to k times the per-district quota receives k seats; a county with population less than one quota receives a fractional vote weight equal to its population share of one quota.

The gerrymandering-resistance of this system is complete: county boundaries are not drawn by anyone in connection with congressional elections. The allocation is mechanically determined by population. There is no drawing step that can be manipulated.

4.2 Proportionality in Competitive States

In competitive states — where the county composition produces both Democratic- leaning and Republican-leaning counties in roughly proportional numbers — county representation produces outcomes that are at least as proportional as algorithmic single-member redistricting and often more so. The reason is that county-level vote shares, while not perfectly proportional, tend to be more moderate than single-member district vote shares, because counties are larger and more geographically heterogeneous than the compact single-member districts that maximize sorting.

Paper E.2 finds that in states like Wisconsin, Michigan, Pennsylvania, and Arizona — the quintessential competitive states of the 2010s–2020s — county representation produces seat–vote deviations of 1–2% versus 3–5% for algorithmic single-member redistricting [Della-Libera, 2026c]. This is a meaningful improvement in proportionality that comes without any drawing step.

4.3 Reduced Proportionality in Highly Urbanized States

In highly urbanized states where large metropolitan counties dominate the population, county representation reintroduces the sorting problem in a different form. Cook County (Chicago) would receive approximately 8 House seats and would elect all 8 at-large. Cook County votes approximately 75–25 Democratic; at-large election of 8 seats from a 75–25 county would produce 6 Democratic seats and 2 Republican seats under proportional allocation — much better than the 7–1 outcome possible under single-member sorting, but still not reflective of the 47% Republican county-wide vote that Republican voters outside Chicago proper would prefer to see represented.

Moreover, the at-large representation mechanism itself is less locally accountable than single-member districts. A voter in Evanston, Illinois has no specific representative accountable to Evanston under a Cook County at-large system; she has 8 representatives nominally accountable to all of Cook County's 5.2 million residents. The constituent-representative bond that single-member districts create — even with their sorting pathology — is entirely absent.

4.4 Responsiveness to Population Change

The second cost of county representation is reduced responsiveness to population change. Congressional districts are redrawn after each decennial census, allowing the map to track population shifts within states. County boundaries, by contrast, are fixed by state statute and change rarely. The allocation of House seats to counties would be recomputed after each census, but the geographic boundaries would remain fixed.

This means that county representation automatically adjusts for inter-county migration (a county that grows relative to the state average receives more seats) but cannot adjust for intra-county concentration changes. If the population of a large county becomes more concentrated in its urban core over a decade — as has happened in most major metropolitan areas since 2000 — county representation provides no mechanism to adjust the representation of intra-county geographic communities.

Paper E.2 simulates the 2010–2020 population shifts under county representation and finds that the system tracks inter-state and inter-county changes reasonably well but systematically under-represents the fastest-growing suburban communities within large counties, which are precisely the communities most politically competitive and most in need of representation adjustment [Della-Libera, 2026c].

4.5 The Fractional-Vote Implementation Problem

The institutional novelty of fractional voting is not merely a public-relations challenge; it creates genuine governance complications. A county whose population entitles it to 0.7 House seats receives a representative with 0.7 votes in Congress. If that representative votes on a bill with 217.4 yeas and 216.6 nays (to fabricate an extreme case), the outcome depends on how fractional votes are tabulated. Current House rules do not accommodate fractional votes, and any implementation would require substantial revision of House rules and likely the Rules of the House adopted at the start of each Congress.

The constitutional question is more fundamental: *Wesberry v. Sanders* [U.S. Supreme Court, 1964] requires that congressional districts be as nearly equal in population as practicable. County populations vary by a factor of nearly 20,000 (from Los Angeles County at 10 million to Loving County, Texas at approximately 64), a variation that is the opposite of "as nearly equal as practicable." The only way to reconcile county representation with equal population is through fractional voting, which has no constitutional precedent in the House of Representatives.

5 Lesson 4: Crossing State Lines Produces Compact but Fragmented Districts

National redistricting without state lines (Paper E.3) is the purest implementation of geometric optimization: treat the continental United States as a single graph of approximately 73,000 census tracts and partition it into 435 compact contiguous districts using the METIS minimum-cut algorithm, without any constraint that district boundaries respect state lines [Della-Libera, 2026d]. The design lesson from this experiment is a precise quantification of the federalism compactness cost: the state-boundary constraint sacrifices approximately 12% of achievable mean Polsby–Popper compactness in exchange for state-based representation.

5.1 The Compactness Gain

The 50-state baseline DIA achieves a mean Polsby–Popper score of 0.367. The national redistricting experiment achieves 0.411, an improvement of $\Delta\text{PP} = +0.044$, or 12.0% of the baseline value [Della-Libera, 2026d].

This improvement is not uniform across the country. It is concentrated in states where state boundaries happen to bisect naturally compact geographic units: the Appalachian ridge that splits the Virginia–West Virginia–North Carolina border region; the Mississippi River basin that straddles multiple state lines in the upper Midwest; the suburban sprawl of metropolitan areas that cross state boundaries (Kansas City, Philadelphia, Cincinnati, Washington DC). In these areas, removing the state constraint allows the algorithm to draw districts that follow natural geographic contours rather than arbitrary political lines.

Approximately 60% of districts are unchanged or minimally changed by removing the state constraint: they would fall entirely within one state under any reasonable compact-district drawing. The compactness gain is concentrated in the remaining 40%, which gain an average of $\Delta\text{PP} = +0.110$ — a substantial improvement in the districts where the state constraint was binding.

5.2 The County Preservation Cost

The compactness improvement comes at a significant cost in county preservation. The baseline DIA preserves 38% of counties whole (no tract splits across district lines). The national redistricting experiment preserves only 23% of counties whole — a 40% reduction in county-preservation rate.

This is counterintuitive at first: if the algorithm is optimizing for compactness without state-line constraints, why does county preservation decline? The answer is that state boundaries often happen to coincide with county boundaries — state lines commonly run along county borders, especially in the South and Midwest where counties were established contemporaneously with state boundaries. When the algorithm is free to cross state lines, it draws districts that cross county lines in new places, fragmenting counties that had been preserved under the state-constrained baseline.

The tradeoff between compactness and county preservation is thus partially mediated by the state-boundary constraint: state boundaries impose compactness costs in some places while incidentally protecting county integrity in others. Removing the state constraint improves geometric compactness while degrading county coherence.

5.3 Political Fragmentation

Beyond county preservation, national redistricting produces a specific type of political fragmentation that does not arise in the state-constrained baseline: districts that cross state lines represent constituents governed by different state laws.

A district spanning the Kansas–Missouri border would represent residents of two states with different income tax rates, different Medicaid eligibility thresholds, different state legislative districts, different US Senate representation, and different electoral calendars. The House member from such a district would face constituent questions — about state programs, state regulations, state courts — that do not map cleanly onto her congressional role. She would also face a reelection challenge defined by two different state primary calendars and two different state party structures.

Paper E.3 identifies 87 cross-state districts in the national map — districts that draw at least 10% of their population from a different state than the majority of their population [Della-Libera, 2026d]. These 87 districts represent a substantial fraction of the 435 total (20%) and are concentrated in the densely settled Northeast and the suburban Midwest.

5.4 The Federalism Cost: A Precise Measurement

The central contribution of E.3 to the design lessons is that it makes the federalism cost of redistricting legible. Before this experiment, the cost of respecting state boundaries in redistricting was unmeasured; the conventional wisdom was simply that state-based redistricting is constitutionally required and therefore not worth analyzing as a design choice.

The finding that the federalism constraint costs approximately 12% mean compactness is both modest enough to be acceptable and substantial enough to be worth knowing. It is modest in the sense that the 0.367 baseline is already substantially higher than the compactness of current commission-drawn maps (which average approximately 0.28–0.31 in most states); the 12% improvement would raise the algorithmic baseline to 0.411, an improvement at the margin. It is substantial in the sense that 12% of mean PP corresponds to visibly more regular district shapes in the cross-state border regions where the improvement is concentrated.

The implication for reform advocates is that the compactness argument for national redistricting is relatively weak: a 12% improvement is not transformative. The proportionality argument — that national redistricting might reduce the sorting-driven seat gap by drawing districts that cross state-level sorting concentrations — is potentially stronger but is not supported by E.3's empirical results (see Lesson 1 above).

6 Lesson 5: The Compactness–Proportionality Pareto Frontier Is Steep

The fifth and most technically precise lesson from the E-track experiments is that the Pareto frontier between compactness and proportionality is steep: a one percentage point improvement in seat–vote proportionality costs approximately 0.015 units of mean Polsby–Popper, roughly 4% of the baseline value. This is the median exchange rate across states; the rate varies by state (range: 0.011–0.023 PP/pp), reflecting differences in urban concentration. We report the median value 0.015 as the headline figure throughout this paper for consistency with the E.0 cross-system synthesis [Della-Libera, 2026a]. No feasible single-election geographic system escapes this trade-off.

6.1 Deriving the Exchange Rate

Paper E.5 implements a parametric optimization that trades compactness for proportionality by varying the weight on the efficiency-gap criterion in the METIS objective function [Della-Libera, 2026f]. At weight = 0 (pure compactness), the algorithm produces the baseline DIA map with mean PP = 0.367 and seat–vote deviation = 4.2 pp. At weight = 1 (pure proportionality), the algorithm produces a map with mean PP = 0.294 and seat–vote deviation = 0.3 pp. The relationship is approximately linear across the weight range, giving a *within-system* exchange rate of:

$$\frac{\Delta\text{PP}}{\Delta\text{deviation}} \approx \frac{0.367 - 0.294}{4.2 - 0.3} = \frac{0.073}{3.9} \approx 0.019 \text{ PP per pp.}$$

This within-system rate (0.019) measures the slope of the E.5 optimization trajectory as a single algorithm trades compactness for proportionality. The E.0 synthesis [Della-Libera, 2026a] estimates a *cross-system* rate of 0.015 PP per pp, measured as the average slope between the DIA baseline and the alternative system points in the Pareto table (multi-member, E.5, etc.). These two numbers measure different phenomena. The cross-system rate (0.015) is the appropriate headline figure for this paper because it characterizes the frontier across the full portfolio of structural alternatives, not only the within-algorithm parametric trajectory. We therefore use 0.015 throughout.

6.2 What the Steep Frontier Means

An exchange rate of 0.015 PP per pp of proportionality improvement is steep because the baseline PP of 0.367 is already substantially higher than most real-world maps. The practical implication is:

- Moving from 4.2 pp seat–vote deviation (baseline) to 2.0 pp deviation (a reform that many proportionality advocates would consider modest) costs approximately $2.2 \times 0.015 = 0.033$ PP, reducing mean PP from 0.367 to 0.334 — a meaningful loss of compactness.
- Achieving 0.5 pp deviation (near-perfect proportionality) costs approximately $3.7 \times 0.015 = 0.056$ PP, reducing mean PP to 0.311 — close to the quality of commission-drawn maps that reformers criticize as insufficiently compact.

In other words, algorithmic proportionality reform would, at the level required to achieve near-perfect seat–vote symmetry, produce maps no more compact than the commission-drawn maps that algorithmic redistricting is intended to improve upon. This is a significant design constraint that proportionality-focused reform proposals must address.

6.3 Why No System Escapes the Frontier

The steep Pareto frontier is a consequence of geographic sorting: the same geographic sorting that produces the seat–vote gap also makes it geometrically expensive to fix. To achieve proportionality, the algorithm must draw districts that crack urban Democratic concentrations; cracking concentrations requires elongated, irregular district shapes; irregular shapes reduce Polsby–Popper. These are consequences of the same underlying geographic distribution of partisan voters, not of the drawing algorithm. We verify this is not an algorithm artifact by confirming the same pattern holds under both edge-weighted bisection (B.2) and standard bisection (B.1) configurations: the exchange rate under standard bisection (0.016 PP/pp) is statistically indistinguishable from the 0.015 figure under edge-weighted bisection, consistent with the sorting hypothesis that the frontier is a property of voter geography rather than of the specific optimization objective.

This can be demonstrated by a thought experiment. Suppose we had a perfectly unbiased redistricting algorithm (if such a thing were possible) that drew districts from a random partition of the census-tract graph subject only to population equality and contiguity. Such an algorithm would produce maps with mean PP near 0.15–0.20 (the compactness of random planar graph partitions) and seat–vote deviations near zero (because partisan sorting would be broken up randomly). Both the compactness and the proportionality of the baseline DIA are improvements over this random baseline, but they are improvements on different objectives, and improving further on one requires accepting deterioration on the other.

The key insight is that the sorting constraint is not a feature that can be "engineered out" of a geographic district system by choosing a different algorithm. It is a property of the data — of where voters live — that any geographically coherent drawing must contend with.

6.4 Implications for Reform Evaluation

The steepness of the Pareto frontier implies that reform proposals should be evaluated not just on the proportionality they achieve but on the compactness they sacrifice. A proposal that achieves 2 pp seat–vote deviation at a cost of 0.033 PP may be acceptable; a proposal that achieves 0.5 pp deviation at a cost of 0.056 PP may not be.

This framework suggests a useful reformulation of the redistricting criteria debate. Rather than treating compactness and proportionality as competing criteria whose relative weights are determined by legislative or judicial preference, we can treat the Pareto frontier itself as an objective criterion: *any redistricting plan should lie on or near the Pareto frontier for its state's geography.* A plan that is less proportional *and* less compact than what the frontier permits is objectively inferior. The debate about where on the frontier to locate the plan — how to trade compactness for proportionality — is a normative question, but it should be conducted with full knowledge of what the trade-off actually costs.

The E-track research program provides the empirical foundation for this frontier-based evaluation. The exchange rates documented in E.5 and corroborated across E.1–E.5 give reformers and courts a quantitative basis for assessing whether a proposed redistricting plan is making a legitimate normative trade-off or simply producing a map that is both less compact and less proportional than the state's geography permits.

7 Synthesis: Design Principles for Geographic Representation Systems

The five lessons from the E-track experiments support a set of design principles that apply beyond the specific systems studied. This section synthesizes those principles and derives their implications for the constraint hierarchy that governs single-member district drawing.

7.1 The Three Design Tensions

The E-track experiments make visible three fundamental tensions in geographic representation system design that are present in all systems but visible only when systems are compared systematically:

Tension 1: Compactness versus Proportionality. The most pervasive tension is between geographic compactness and partisan proportionality. Lesson 5 quantifies this tension at the Pareto frontier: 0.015 PP per pp of proportionality gained. Lesson 1 explains its cause: geographic sorting makes Democrats geographically inefficient in single-member winner-take-all systems. Lesson 2 establishes that multi-member districts partially but not fully relieve this tension by averaging over more geographic heterogeneity.

This tension is not merely a feature of the US context. Rodden [2019] documents analogous sorting patterns in the UK, Canada, and Australia. Paper E.6 finds the same proportionality–compactness exchange in all four Westminster-system countries studied, confirming that the tension is a property of the geographic sorting–winner-take-all combination, not of US partisan geography specifically.

Tension 2: Community Preservation versus Compactness. Lesson 3 and Lesson 4 both surface a second tension: the desire to preserve pre-existing political communities (counties, states) conflicts with geometric compactness optimization. State boundaries cost 12% PP (Lesson 4); county preservation at the 85% level costs 18% PP (from the E.2 results [Della-Libera, 2026c]). These costs are independent: preserving both counties and states costs approximately 28% PP relative to unconstrained national optimization.

The community-preservation tension is normatively distinct from the compactness–proportionality tension. Proportionality is contested as a redistricting goal (courts have declined to constitutionalize it); community preservation is widely accepted as a legitimate goal (most state redistricting statutes list it as a criterion). This means that community-preservation costs are costs that the political system has implicitly agreed to pay, whereas compactness–proportionality trade-offs are costs that are being debated.

Tension 3: Structural Design versus Drawing Optimization. The deepest tension is not between two objectives on the Pareto frontier but between two different levels of reform. Lessons 1 and 2 together establish that within any single-member geographic system, drawing optimization (algorithmic redistricting, independent commissions, court supervision) can eliminate process-level distortions but cannot eliminate structure-level distortions arising from geographic sorting. Structural reform — multi-member districts — is required to materially change the seat–vote relationship.

This tension implies that the redistricting reform debate has been conducted partly at the wrong level. The bulk of litigation, legislative effort, and research attention has focused on drawing-process reform. The structural reform that would actually change the seat–vote relationship (multi-member districts) has received far less policy attention and far less empirical evaluation. The E-track experiments are intended partly to correct this imbalance by providing rigorous comparative evidence on what structural reform achieves.

7.2 The Optimal Constraint Hierarchy for Single-Member Systems

Given these three tensions, what is the optimal constraint hierarchy for single-member district drawing? We argue that the current practice — population equality first, contiguity second, compactness third, with community preservation as a secondary criterion — is optimal for the following reasons.

Population equality first is constitutionally mandated (*Wesberry* [U.S. Supreme Court, 1964]) and electorally foundational. No other criterion can justify systematic population inequality.

Contiguity second follows from population equality: a non-contiguous district is geometrically incoherent as a representation unit and creates representational obligations that the representative cannot fulfill for constituents with no geographic connection to the district's main body.

Compactness third follows from contiguity: given population equality and contiguity, the shape of the district is the primary remaining degree of freedom. Compactness optimization within this constraint yields the districts most resistant to partisan manipulation [Polsby and Popper, 1991] and most geographically coherent as communities of representation.

Community preservation as secondary criterion reflects the lesson that community preservation is valuable but costs compactness. Treating it as a hard constraint (preserve all counties) is too expensive; treating it as a soft criterion (preserve counties where doing so does not substantially reduce compactness) is the right balance.

The key design lesson for single-member systems is that adding proportionality as a criterion at any level of the hierarchy is inappropriate: it substitutes a structural problem (geographic sorting) for a drawing problem, and attempts to solve the structural problem through drawing will produce maps that are less compact than what the geography permits without achieving the proportionality goal reliably.

7.3 The Case for Multi-Member Districts as Structural Reform

The synthesis of the five lessons points consistently toward multi-member districts as the only mechanism within a geographic, single-election representation system that can materially improve both proportionality and minority representation without sacrificing the constitutional advantages of district-based representation (geographic coherence, constituent-representative accountability, local focus).

The advantages of multi-member districts over single-member alternatives are:

1. **Partial sorting escape**: 40% gap reduction (three-member) to 65% gap reduction (five-member), compared to 0% for any single-member drawing method (Lessons 1 and 2).

2. **Minority representation improvement**: +18% majority-minority seats without requiring geographic concentration (Lesson 2).

3. **Constitutional availability**: Requires only federal statute to repeal the 2 U.S.C. §2c single-member mandate (E.0 constitutional taxonomy).

4. **Compactness compatibility**: Three-member districts achieve most of the proportionality gain at a compactness cost (22% PP reduction) that is large but not catastrophic — far smaller than the 34%–56% cost of five-member or pure-proportionality systems.

The disadvantages — ballot complexity, weaker local ties, incumbent advantage in large districts — are real but are normative objections rather than empirical disqualifiers. The ballot-complexity objection is partially addressed by ranked-choice voting experience in STV systems (Ireland, Australia), where voter error rates are modest and declining [Farrell, 2011].

We do not advocate for multi-member districts as a policy position. We observe that the E-track research program, taken as a whole, provides a stronger empirical foundation for the structural case for multi-member districts than previously existed, and that this evidence should inform reform debates even if normative judgments about ballot complexity and local accountability ultimately determine the political outcome.

8 Conclusion

The six experiments in Track E constitute the most systematic algorithmic exploration of the congressional representation design space to date. By holding the core algorithm constant and varying design dimensions in controlled ways, the experiments produce comparable, quantifiable evidence about the costs and benefits of structural alternatives to single-member geographic districting.

The five lessons we derive from this evidence are:

1. **Geography is destiny for single-member districts.** The Rodden sorting gap is not a drawing artifact. No single-member geographic drawing method — algorithmic, commission-based, or judicially supervised — eliminates systematic underrepresentation of Democrats relative to national vote share. The cause is where Democratic voters live, not how district boundaries are drawn.

2. **Multi-member districts partially but not fully escape sorting.** Three-member districts reduce the gap by 40%; five-member by 65%. The residual gap reflects extreme urban concentration that no realistic district size can dilute given state-boundary constraints. Multi-member districts also improve minority representation by 18% without requiring geographic concentration.

3. **Eliminating districts trades proportionality for local accountability.** County representation improves proportionality in competitive states but introduces fractional voting, reduces constituent-representative bonds, and is constitutionally unavailable in its strict form without amendment.

4. **Crossing state lines produces compact but fragmented districts.** National redistricting gains 12% mean Polsby–Popper but loses 40% of county preservation and creates 87 cross-state districts that fragment state-based political accountability. The federalism cost is real but modest; the representation fragmentation cost is harder to quantify but significant.

5. **The compactness–proportionality frontier is steep.** Gaining 1 pp of proportionality costs 0.015 PP units. Achieving near-perfect proportionality requires maps no more compact than the commission-drawn maps that algorithmic redistricting is intended to improve. No feasible single-election geographic system escapes this trade-off.

These lessons support a clear design hierarchy. For single-member systems operating within current constitutional constraints, the optimal constraint ordering is population equality, contiguity, and compactness — with community preservation as a soft secondary criterion. Proportionality should not be added as a drawing criterion within single-member systems because it addresses a structural problem through a drawing instrument and produces inferior results on both compactness and proportionality relative to structural reform.

The structural reform that the evidence most clearly supports is multi-member districts with proportional allocation via ranked-choice voting. This reform is constitutionally available without amendment, produces material improvement in both proportionality and minority representation, and is the only mechanism within a geographic representation system that partially escapes the Rodden sorting constraint. Its costs — ballot complexity, weaker local ties — are real but normative rather than empirical, and the E-track evidence provides the empirical foundation on which normative debates should proceed.

8.1 Limitations and Future Research

This paper draws on simulated algorithmic outcomes rather than actual electoral results. The vote shares used to compute proportionality deviations are historical averages; future elections may differ. The geographic data are from the 2020 Census; demographic and residential sorting patterns will change over time. The sorting-gap estimates are derived from House election results and may not generalize to other office levels.

Future research should examine hybrid systems — multi-member districts in urban areas combined with single-member districts in rural areas — which are not included in the E-track experiments. Such hybrid systems might achieve a better proportionality–compactness balance than either pure multi-member or pure single-member systems, since they concentrate the multi-member mechanism where it provides the most benefit (urban sorting concentrations) while preserving the local-accountability advantages of single-member districts where they are most valued (rural and small-town constituencies).

The international evidence (E.6) suggests that algorithmic redistricting provides compactness improvements in all Westminster-system countries studied. Future research should extend this analysis to mixed-member proportional systems (Germany, New Zealand) and proportional representation systems (Nordic countries) to test whether any of the lessons derived from the single-member US context apply in fundamentally different electoral architectures.

8.2 Coda: The Value of Algorithmic Counterfactuals

The E-track research program demonstrates that algorithmic implementation of constitutionally unavailable systems (national redistricting) and politically controversial systems (partisan-similarity clustering) has significant analytical value even when the systems cannot be adopted. The ability to implement and measure the outcomes of counterfactual systems is a uniquely algorithmic contribution to the representation design literature. Manual redistricting cannot implement national redistricting (it is unconstitutional), cannot implement strict partisan-fairness allocation (it is legally prohibited), and cannot implement county representation at scale (it has never been attempted in the US context). Algorithmic redistricting can implement all of these as measurement instruments, producing quantitative evidence about what we pay for the constitutional and institutional choices we have made.

The lesson for political science methodology is that algorithmic implementation of counterfactual institutions is a powerful tool for causal inference about representation system design. The lesson for redistricting reform is that the evidence thus produced should inform the normative debates that determine which institutional choices we make going forward.

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